Project Report

Spring 2025

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Project: Vehicle Sales

Note: As you write each section, try to be as clear and detailed as possible. Your goal is to communicate your thought process and work clearly. Don't worry if you faced challenges or made mistakes; discussing these is a valuable part of learning and shows your problem-solving skills! Remember, there's no single 'right' way to do these tasks, so be creative and honest in your responses.

Problem Statement (2-3 Paragraphs):

First, you try to read a file of car sales, then let the user pick menu options to see the totals. The program gives back annual and monthly tablets for either all brands or one brand's models, matching the format my teacher expects.

Stuff can go wrong if the file is missing, broken, or if someone runs reports before loading the data. It catches file errors, makes sure if the header has Brand/Model plus all the months, warns the user to import first so nothing crashes. To cover weird cases, I changed monthly numbers, automatically made empty ones to zero, and skipped any row. If the header was missing one of its months, the importer has a friendly message instead of printing an error message.

Design (1-3 Paragraphs):

I split things into smaller files and share interfaces so the table printer can run them the same. The data manager reads the file into a dictionary and displays utils prints everything. In main, I changed the menu so all the similar options call the same helper, which keeps logic short. The shared interface means the table printer doesn't care if a row represents a model or brand, it just asks for names and numbers to print. I streamlined all the similar menu choices through show_all_brands to avoid printing stuff over and over again.

Testing (1-2 Paragraphs + screenshots of 3 test cases):

I ran the script, imported the real file, and tried other menu options to make sure the tables line up. I also typed wrong choices, tried reports before importing, and entered unavailable brands. Everything worked as expected. After each change, I reran the program, hit option 1 to import, then went through all the options to make sure everything was working as expected.

```
Choose an action from this list by its number:
1 - Import Vehicle Sales Data
2 - Display annual Sales Data for all Brands
3 - Display monthly Sales Data for all Brands
4 - Display annual Sales Data by Model for one Brand
5 - Display monthly Sales Data by Model for one Brand
0 - Exit
Select an action by its number: (0-5) 4

Select a brand from this list by name or number:
 1 - Acura
 2 - Alfa Romeo
 3 - Audi
 4 - Bentley
 5 - BMW
 6 - Buick
 7 - Cadillac
 8 - Chevrolet
 9 - Chrysler
10 - Dodge
11 - Ford
12 - Genesis
13 - GMC
14 - Honda
15 - Hyundai
16 - Infiniti
17 - Jaguar
18 - Jeep
19 - Kia
20 - Land Rover
21 - Lexus
22 - Lucid
23 - Mazda
24 - Mercedes-Benz
25 - Mini
26 - Mitsubishi
27 - Nissan
28 - Polestar
29 - Porsche
30 - Ram
31 - Rivian
32 - Subaru
33 - Tesla
34 - Toyota
35 - Volkswagen
36 - Volvo
Choose a brand by entering its name or number: Foorrd
Brand not found. Try again.
Choose a brand by entering its name or number:
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HW3 - Vehicle Sales
Solution by Rakshith Jayakarthikeyan

Choose an action from this list by its number:
1 - Import Vehicle Sales Data
2 - Display annual Sales Data for all Brands
3 - Display monthly Sales Data for all Brands
4 - Display annual Sales Data by Model for one Brand
5 - Display monthly Sales Data by Model for one Brand
0 - Exit
Select an action by its number: (0-5) 1

Loaded 209 models across 36 brands.

Choose an action from this list by its number:
1 - Import Vehicle Sales Data
2 - Display annual Sales Data for all Brands
3 - Display monthly Sales Data for all Brands
4 - Display annual Sales Data by Model for one Brand
5 - Display monthly Sales Data by Model for one Brand
0 - Exit
Select an action by its number: (0-5) 2

Annual Sales for All Brands

Brand         Model             Annual Sales
Acura                                 134,751
Alfa Romeo                              5,655
Audi                                  140,547
Bentley                                 3,207
BMW                                    89,945
Buick                                 125,527
Cadillac                              106,419
Chevrolet                             391,550
Chrysler                              106,237
Dodge                                  77,915
Ford                                  819,544
Genesis                                88,462
GMC                                   615,927
Honda                               1,281,911
Hyundai                               766,802
Infiniti                              116,453
Jaguar                                 17,123
Jeep                                  476,777
Kia                                   589,193
Land Rover                            138,261
Lexus                                 672,110
Lucid                                  14,233
Mazda                                 294,969
Mercedes-Benz                         169,163
Mini                                   52,768
Mitsubishi                            169,077
Nissan                                589,090
Polestar                               26,808
Porsche                               147,323
Ram                                   446,824
Rivian                                 95,642
Subaru                                343,501
Tesla                                 544,502
Toyota                              2,002,138
Volkswagen                            271,244
Volvo                                  92,437
```

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Choose an action from this list by its number:
1 - Import Vehicle Sales Data
2 - Display annual Sales Data for all Brands
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4 - Display annual Sales Data by Model for one Brand
5 - Display monthly Sales Data by Model for one Brand
0 - Exit
Select an action by its number: (0-5) 3

Data not loaded. Use option 1 first.
```

Conclusion (1 paragraph)

The project does what the assignment asked. I'd call it a good project because every feature works as intended. Next time I'd do the same modular setup but add a couple automated tasks to make it more efficient.